Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

HHS Public Access

Author manuscript

Neurourol Urodyn. Author manuscript; available in PMC 2020 September 01.

Published in final edited form as:

Neurourol Urodyn. 2019 September ; 38(7): 1828–1833. doi:10.1002/nau.24108.

Neural control properties of the external anal sphincter in young and elderly women.

Nicholas Dias[1], **Chuan Zhang**[1], **Xuhong Li**[2], **Leila Neshatian**[3], **Francisco J Orejuela**[4], **Yingchun Zhang**[1,*]

[1]Department of Biomedical Engineering, University of Houston, Houston, TX, USA;

[2]The Third Xiangya Hospital; Central South University, Changsha, China;

[3]Gastroenterology & Hepatology, Stanford University School of Medicine, Stanford, CA, USA;

[4]Obstetrics and Gynecology, Baylor College of Medicine, Houston, TX, USA.

Abstract

Aims: The prevalence of fecal incontinence (FI) increases with age and affects more than 15% of the elderly population. Sarcopenia, skeletal muscle structural and functional decline with aging, is known to be caused by neuromuscular dysfunction. However, age-related alterations of the neuromuscular function of the external anal sphincter (EAS) have not been studied. This study aims to quantitatively characterize the effect of aging on the EAS by assessing the firing patterns and size of motor unit action potential (MUAP) using high-density surface EMG (HD-sEMG) recording and analysis techniques.

Methods: 13 young (31.0±3.6 years) and 14 elderly (64.3±6.2 years) healthy women were recruited for this study. EMG activity of the EAS during maximal voluntary contraction was recorded by a 64-Channel, HD-sEMG intra-rectal probe. HD-sEMG signals were decomposed into MUAP spike trains to extract the firing rate and amplitudes thereof.

Results: HD-sEMG decomposition was successfully performed. For the young and elderly groups, mean motor unit (MU) firing rates of 11.4±2.1 pulses per second (PPS) and 9.6±2.3 PPS, and mean MUAP amplitudes of 45.2±14.3 μV and 61.9±21.2 μV, were respectively obtained. Both the MU firing rate, and MUAP amplitude were significantly different between two groups ($p<.05$). Moreover, the MUAP firing rate and amplitude correlated with age with a linear regression model ($p<.05$).

Conclusions: This study represents the first effort to examine the effect of aging on the neuromuscular function of EAS. Results suggest an age-related impairment of lower motor neuron descending excitation to the EAS with a compensatory increase in mean MU size.

*Corresponding author: Yingchun Zhang, Ph.D., Department of Biomedical Engineering, University of Houston, 4849 Calhoun Rd., Rm 373, Houston, TX 77004 USA, yzhang94@uh.edu, Telephone: +1-713-743-6127.

Ethics approval and consent to participate

The study protocol was approved by the local Institutional Review Board (IRB), and written consents were obtained from the participants before the experiments.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Keywords

External Anal Sphincter; Multi-channel Electromyography; Motor Unit Action Potential; Motor Unit Control

INTRODUCTION

Fecal incontinence (FI), defined as the involuntary escape of stool, is a devastating anorectal disorder that negatively impacts the quality of life of the affected individuals. The external anal sphincter (EAS), part of the levator ani, is essential for maintaining continence at rest and during an increase in intra-abdominal pressure.[1] The prevalence of FI increases with age,[2] from 2.6% at ages 20–30 to more than 15% in people >70 years old[3] Epidemiological studies have also demonstrated age-related reduction in the anal squeeze pressures which may predispose elderly individuals to FI.[3,4] Although the exact pathophysiology of age-related changes in EAS function is not fully understood, EAS muscle atrophy and sarcopenia with associated functional impairment is suggested as the potential cause of these changes.[5]

The integrity of the neuromuscular system, which involves the skeletal muscles and motor neurons and the neuromuscular junctions determines its functional capacity. Aging is associated with impairment of different components of the neuromuscular system, which ultimately results in loss of muscle mass and function. The skeletal muscle functional and structural decline with aging, sarcopenia has been widely studied in the skeletal muscles of extremities, concluding that an up to 40% loss in muscle force could be associated with aging.[6,7]. This loss has been attributed to alterations in motor unit properties during the normal aging process, and has been extensively studies across the different muscle groups.[8–10] Notably, most of these studies observed aging-related alterations that include reductions in mean motor unit firing rate. Moreover, Kamen et. al reported a significant decrease in motor unit firing rate under maximal voluntary contraction.[8] Additionally, Stalberg et. al and de Koening et. al both reported increases in macro electromyography (EMG) amplitude in the quadriceps femoris and tibialis anterior muscles with increased age, respectively.[11,12] Despite the effort to characterize the motor unit properties in sarcopenia in a variety of skeletal muscles, there is a lack of understanding of the role of sarcopenia in levator ani muscle function, possibly due to the complicated anatomy of the pelvis making direct access difficult.[13]

Motor unit properties are commonly assessed electro-physiologically with intramuscular EMG (iEMG). Unfortunately, iEMG is unable to access more than a few motor units per insertion, necessitating repeated invasive, painful sampling with relatively low test-retest repeatability. The current non-invasive alternative, conventional surface EMG, underestimates activation sent from the spinal cord due to amplitude cancellation of the positive and negative phases of motor unit action potentials (MUAPs).[14] Anorectal ultrasound and manometry can assess the anatomical and functional health of the EAS, but do not provide any insight into the underlying pathophysiology. Recent advances in high-density surface EMG (HD-sEMG) have yielded promising solutions towards the non-

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

invasive, reliable extraction of global motor unit firing information, muscle innervation, and single MUAP analysis.[13,15–18] Compared to iEMG, HD-sEMG holds a significant advantage in that a single session can yield detailed information from an abundance of broadly distributed motor units, thus offering a global picture of the constitutive motor unit properties of a muscle, rather than a few motor units within the uptake area of the iEMG sensor.

Our intra-rectal HD-sEMG probe has been designed to provide close muscle contact and complete coverage of the EAS while minimizing patient discomfort. The broad spatial coverage allows for successful and reliable EMG decomposition.[13,16] In this study, for the first time, we employed our non-invasive intra-rectal HD-sEMG recording and analysis tools to study the age-related alterations of the motor unit properties and MUAP size of the EAS in healthy women. The quantitative changes in EAS skeletal muscle during the natural aging is also of significant clinical importance and could translate into specific intervention strategies, resulting in limiting deterioration of muscle function, and alleviating the levator ani sarcopenia.

MATERIALS AND METHODS

Healthy women were recruited to study the age-related alterations via EAS electrophysiology. A detailed health questionnaire was used to collect subjects' demographics, and medical history, including obstetric history, the presence of anal incontinence, and/or neuromuscular disorders. Exclusion criteria were reported involuntary loss of feces over the preceding three-month period, history of neuromuscular disorders, history of obstetric EAS injury, and/or pelvic surgery. All subjects were further screened via transperineal ultrasound, and subjects with identifiable perianal or EAS scarring were excluded from participation to avoid confounding factors. All ultrasound and HD-sEMG tests were performed after obtaining written consent by an experienced physician with an approved study protocol reviewed by the local Institutional Review Board (IRB).

Subjects were situated in the lithotomy position and were trained on how to consistently squeeze their EAS prior to the experiment. A thoroughly soaked Velcro strap ground was attached to the right wrist, and a self-adhesive reference electrode was fixed to the inner thigh. A 64-channel HD-sEMG probe with a length of 175 mm and a diameter of 14.4mm was lubricated with conductive gel and inserted to the rectal space. Each EMG sensor features a rectangular shape with an axial length of 4.0 mm and a width of 2.4 mm. Consideration was given to ensure that consistent probe orientation and depth was maintained during the test. The subject was then instructed to squeeze her EAS and maintain the contraction for 10 seconds with maximal effort. This contraction was repeated six times and an adequate rest interval was given to avoid muscle fatigue. HD-sEMG signals were amplified through a Brainvision amplifier (Brain Products, Gilching, Germany) at a sampling frequency of 5,000 Hz, and stored for offline signal processing.

Raw EMG data were bandpass filtered from 10 Hz to 400 Hz and notch filtered at 50 Hz with second-order Butterworth filters using MATLAB 2015a (Mathworks Inc., Natick MA). Contraction periods were segmented with respect to the markers generated during signal

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

acquisition. A k-means clustering convolution kernel compensation (KmCKC) method was employed to decompose surface interferential EMG signals into constituent MUAP trains.[19] Corresponding MUAP waveforms were reconstructed following the spike-triggered averaging (STA) method. Firing rates for each motor unit spike train were calculated as the average number of motor unit firings per second during the most consistent contraction period. The motor unit firing rate for each subject was defined as the mean firing rate of all decomposed motor units. The innervation zone channel was visually identified from the reconstructed bipolar mapping by inspecting the reversal of signal polarity[13,16]. The MUAP size was estimated as the largest negative peak in the reconstructed motor unit waveform that presents above the innervation zone.

Comparisons of the firing rates and motor unit amplitude between the young and elderly group were evaluated by a two-sample student's t-test. A linear regression model was also employed to evaluate the relationship between age and the two motor unit features defined. Statistical significance was assessed with MATLAB 2015a, and accepted at $p<0.05$.

RESULTS

A total of 27 women were enrolled in this study, with 13 young (31.0±3.6 years) and 14 elderly (64.3±6.2 years). One subject from the elderly group was excluded from analysis due to poor signal quality. Table 1 summarizes demographic information for both groups. Figure 1a shows the study paradigm with a representative signal acquired from one channel during a voluntary squeezing maneuver.

HD-sEMG decomposition was successfully performed in all the subjects tested. On average, decomposition yielded 7.2±2.9 motor units (MUs) for the young subjects and 7.2±2.3 MUs for the elderly. A set of decomposed spike trains from one subject is shown in Figure 1b. The mean firing rate was significantly higher in the young group at 11.4±2.1 pulses per second (PPS) as compared to the elderly group at 9.6±2.3 PPS ($p<0.05$). HD-sEMG signals were averaged with respect to the decomposed spike times to give a 64-channel MUAP profile for each MU acquired from each probe, shown as Figure 2a. Figure 2b shows the STA method for one row of signals. This process was repeated for all MUs and averaged across all subjects. MUAP amplitudes of 45.2±14.3 μV and 61.9±21.2 μV were obtained for the young and elderly groups, respectively. This was statistically significant ($p<0.05$).

By fitting to a linear regression model, we showed that the firing rate was negatively correlated with age ($p<0.05$); Figure 3. Moreover, MUAP size correlated with age ($p<0.05$); Figure 4.

DISCUSSION

In this study, we aimed to quantitatively assess the age-related neuromuscular changes of EAS determined by intra-rectal HD-sEMG recording and decomposition techniques. We employed HD-sEMG with small contact patches and inter-electrode distances to record vast spatiotemporal information across the EAS, which in turn allows for detailed neuromuscular information to be extracted at a resolution comparable to intramuscular EMG.[20] Our results

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

evidenced age-related alterations of the motor unit firing and morphological behavior occur in tandem, resulting in EAS sarcopenia.

Aging has been shown to alter the local innervation remodeling at the neuromuscular junction. The gradual loss of anterior horn cells and degeneration of spinal motor neurons leave the muscle fibers denervated. Neighboring motor neurons then undergo a compensatory “sprouting” process which increases the innervation ratio (muscle fiber number per motor unit) and enlarged MUAP.[21] Histological data from rats has shown up to a 300% increase in Type I fiber ratio with aging.[22] Electrophysiological studies have demonstrated that older individuals exhibit larger motor units than younger individuals in other skeletal muscles[11,12] Our results confirm the presence of the same remodeling process explained above in the EAS, using a non-invasive approach. MUAP amplitude was significantly higher in the elderly group ($p<.01$), as shown in Figure 4. A linear regression model further suggested that MUAP amplitude may be linearly correlated with advancing age ($p<.05$), although a future study with increased sample size is needed to confirm this postulation. Our results are consistent with reported single fiber EMG studies, which showed a significant increase in fiber density in subjects older than 60 years of age.[23]

Motor control is modulated by the central excitatory drive and peripheral motor unit recruitment. The activation of motor units is based on a response to a single common drive, i.e. the net combination of CNS and inhibitory proprioceptive feedback, rather than the CNS controlling each motor unit separately.[24] Upon peripheral degeneration, it has been assumed that the firing rates of the remaining motor units must increase to produce the same force. Early studies by Dorfman *et al.* demonstrated that motor unit firing rate increases after motor denervation due to amyotrophic lateral sclerosis (ALS) and spinal muscular atrophy. On the other hand, a marked decrease in motor unit firing rate has been reported from the central neuronal deficit.[25] Kasi *et al.* later found that lower motor neuron dysfunction increased the mean firing rate; and conversely, upper motor neuron dysfunction caused a significant decrease in firing rate.[26] Rose *et al.* found that cerebral palsy sufferers were unable to recruit fast firing motor units, reporting an almost two-fold decrease in maximal motor unit firing rate in the tibialis anterior and medial gastrocnemius, suggesting a major central component to weakness related to cerebral palsy.[27] Taken together, the findings of these studies clearly show that disorders with a central origin reduce the central drive to the lower motor neurons, manifesting in EMG as a reduction motor unit firing rate. Conversely, in disorders where motor units are lost, there is an effort to compensate by increasing the firing rate of the remaining motor units. In other words, firing rate analysis can characterize the level at which motor neuron disturbances occur.

In our study, aging was associated with a significant decrease in EAS firing rates during MVC ($p<.05$), as shown in Figure 3, which suggests a disrupted ability of the central nervous system to modulate muscle contraction. Interestingly, despite clear evidence of motor unit sprouting due to compensatory re-innervation, we found that motor unit firing rate does not increase with age, as would be expected to compensate for the loss of motor units. Similar effects of aging have been reported in the other skeletal muscles[28,29] of healthy subjects and our results confirm that the same reduction in motor unit firing rate exists in the EAS. This can be explained by the dominant impact of diminished upper motor

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

neuron drive to the anterior horn cells. Furthermore, during innervation remodeling, type II muscle fibers are denervated and re-innervated by slow motor units.[30] Consequently, the re-innervated type II fast-twitch fibers begin to approximate type I slow-twitch fibers.[31] Histochemical evidence of this remodeling process has been shown.[32,33] If remodeling does increase the ratio of type I slow-twitch fibers, one might expect a compromised initial "burst" response of the EAS to high-pressure events, producing a pressure environment conducive to leakage, which may partially explain the increased prevalence of FI in the elderly. In fact, prior studies have shown that women with fecal incontinence are less susceptible to EAS fatigue than continent women, which may be due to a higher ratio of Type I fibers in the EAS[34].

By evaluating MUAP firing rate and amplitude with respect to age, our results were expected to further advance our mechanistic understanding of EAS weakness in the elderly. As no concurrent measure of EMG and muscle force can be performed with currently available technology, the study only analyzed motor unit firing patterns at maximal voluntary contractions. Nonetheless, maximal contraction demonstrates the maximum capacity of motor unit recruitment and provides important insight into alterations in motor unit properties and control. In this study, participants were limited to a healthy population to control for the confounding effects of neuromuscular disorders or sphincter trauma. Future studies should aim to characterize the motor unit properties in FI patients. If pelvic floor muscle training is chosen as a treatment strategy, future studies can use HD-sEMG to longitudinally track the neuromuscular changes that occur during training regimens.

CONCLUSIONS

Our results show, for the first time, that a significant decrease in EAS motor unit firing rates exists in the elderly population. Secondly, we report a significant increase in the average size of the EAS MUAPs using a non-invasive HD-sEMG technique, which is consistent with previously reported intramuscular EMG and histochemical results. This relationship suggests that EAS weakness in the elderly may be caused by the parallel denervation of muscle fibers, combined with a diminished upper motor neuron drive to the anterior horn cells. Future studies should focus on employing this information to identify clinical neurophysiological biomarkers to aid the prediction, early diagnosis and management of FI with aging.

ACKNOWLEDGMENTS

This work was supported by NIH DK082644, DK113525, AG053778, and the University of Houston.

REFERENCES

1. Lewicky-Gaupp C, Hamilton Q, Ashton-Miller J, Huebner M, DeLancey JO, Fenner DE. Anal sphincter structure and function relationships in aging and fecal incontinence. American journal of obstetrics and gynecology. 2009;200(5):559. e551–559. e555. [PubMed: 19136096]
2. Quander CR, Morris MC, Melson J, Bienias JL, Evans DA. Prevalence of and factors associated with fecal incontinence in a large community study of older individuals. The American journal of gastroenterology. 2005;100(4):905. [PubMed: 15784039]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

3. Whitehead WE, Borrud L, Goode PS, et al. Fecal incontinence in US adults: epidemiology and risk factors. Gastroenterology. 2009;137(2):512–517. e512. [PubMed: 19410574]

4. Fox JC, Fletcher JG, Zinsmeister AR, Seide B, Riederer SJ, Bharucha AE. Effect of aging on anorectal and pelvic floor functions in females. Diseases of the colon & rectum. 2006;49(11):1726–1735. [PubMed: 17041752]

5. Laurberg S, Swash M. Effects of aging on the anorectal sphincters and their innervation. Diseases of the colon & rectum. 1989;32(9):737–742. [PubMed: 2758941]

6. Porter MM, Vandervoort AA, Lexell J. Aging of human muscle: structure, function and adaptability. Scandinavian journal of medicine & science in sports. 1995;5(3):129–142. [PubMed: 7552755]

7. Goodpaster BH, Park SW, Harris TB, et al. The loss of skeletal muscle strength, mass, and quality in older adults: the health, aging and body composition study. The Journals of Gerontology Series A: Biological Sciences and Medical Sciences. 2006;61(10):1059–1064.

8. Kamen G, Sison SV, Du C, Patten C. Motor unit discharge behavior in older adults during maximal-effort contractions. Journal of Applied Physiology. 1995;79(6):1908–1913. [PubMed: 8847252]

9. Nelson RM, Soderberg GL, Urbscheit NL. Alteration of motor-unit discharge characteristics in aged humans. Physical therapy. 1984;64(1):29–34. [PubMed: 6691051]

10. Howard JE, McGill KC, Dorfman LJ. Age effects on properties of motor unit action potentials: ADEMG analysis. Annals of Neurology: Official Journal of the American Neurological Association and the Child Neurology Society. 1988;24(2):207–213.

11. De Koning P, Wieneke G, Van Spijk DvdM, Van Huffelen A, Gispen W, Jennekens F. Estimation of the number of motor units based on macro-EMG. Journal of Neurology, Neurosurgery & Psychiatry. 1988;51(3):403–411.

12. Stålberg E, Borges O, Ericsson M, et al. The quadriceps femoris muscle in 20–70-year-old subjects: relationship between knee extension torque, electrophysiological parameters, and muscle fiber characteristics. Muscle & Nerve: Official Journal of the American Association of Electrodiagnostic Medicine. 1989;12(5):382–389.

13. Peng Y, He J, Khavari R, Boone TB, Zhang Y. Functional mapping of the pelvic floor and sphincter muscles from high-density surface EMG recordings. International urogynecology journal. 2016;27(11):1689–1696. [PubMed: 27193113]

14. Keenan KG, Farina D, Maluf KS, Merletti R, Enoka RM. Influence of amplitude cancellation on the simulated surface electromyogram. Journal of Applied Physiology. 2005;98(1):120–131. [PubMed: 15377649]

15. Ullah K, Cescon C, Afsharipour B, Merletti R. Automatic detection of motor unit innervation zones of the external anal sphincter by multichannel surface EMG. Journal of Electromyography and Kinesiology. 2014;24(6):860–867. [PubMed: 24948528]

16. Dias N, Li X, Zhang C, Zhang Y. Innervation asymmetry of the external anal sphincter in aging characterized from high-density intra-rectal surface EMG recordings. Neurourology and urodynamics. 2018.

17. Peng Y, He J, Yao B, Li S, Zhou P, Zhang Y. Motor unit number estimation based on high-density surface electromyography decomposition. Clinical Neurophysiology. 2016;127(9):3059–3065. [PubMed: 27472541]

18. Watanabe K, Holobar A, Kouzaki M, Ogawa M, Akima H, Moritani T. Age-related changes in motor unit firing pattern of vastus lateralis muscle during low-moderate contraction. Age (Dordrecht, Netherlands). 6 2016;38(3):48.

19. Ning Y, Zhu X, Zhu S, Zhang Y. Surface EMG decomposition based on K-means clustering and convolution kernel compensation. IEEE journal of biomedical and health informatics. 2015;19(2): 471–477. [PubMed: 25486655]

20. Li X, Holobar A, Gazzoni M, Merletti R, Rymer WZ, Zhou P. Examination of poststroke alteration in motor unit firing behavior using high-density surface EMG decomposition. IEEE Transactions on Biomedical Engineering. 2015;62(5):1242–1252. [PubMed: 25389239]

21. Doherty TJ, Vandervoort AA, Taylor AW, Brown WF. Effects of motor unit losses on strength in older men and women. Journal of Applied Physiology. 1993;74(2):868–874. [PubMed: 8458808]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

22. Kung TA, Cederna PS, van der Meulen JH, Urbanchek MG, Kuzon WM Jr, Faulkner JA. Motor unit changes seen with skeletal muscle sarcopenia in oldest old rats. Journals of Gerontology Series A: Biomedical Sciences and Medical Sciences. 2013;69(6):657–665.
23. Neill M, Swash M. Increased motor unit fibre density in the external anal sphincter muscle in ano-rectal incontinence: a single fibre EMG study. Journal of Neurology, Neurosurgery & Psychiatry. 1980;43(4):343–347.
24. De Luca C, LeFever R, McCue M, Xenakis A. Control scheme governing concurrently active human motor units during voluntary contractions. The Journal of physiology. 1982;329(1):129–142. [PubMed: 7143247]
25. Dorfman LJ, Howard JE, McGill KC. Motor unit firing rates and firing rate variability in the detection of neuromuscular disorders. Electroencephalography and clinical neurophysiology. 1989;73(3):215–224. [PubMed: 2475326]
26. Kasi PK, Krivickas LS, Meister M, et al. Motor unit firing characteristics in patients with amyotrophic lateral sclerosis. Paper presented at: Bioengineering Conference, 2009 IEEE 35th Annual Northeast2009.
27. Rose J, McGill KC. Neuromuscular activation and motor-unit firing characteristics in cerebral palsy. Developmental medicine and child neurology. 2005;47(5):329–336. [PubMed: 15892375]
28. Newton J, Yemm R, McDonagh M. Study of age changes in the motor units of the first dorsal interosseous muscle in man. Gerontology. 1988;34(3):115–119. [PubMed: 3417150]
29. Connelly DM, Rice CL, Roos MR, Vandervoort AA. Motor unit firing rates and contractile properties in tibialis anterior of young and old men. Journal of Applied Physiology. 1999;87(2):843–852. [PubMed: 10444648]
30. Roos MR, Rice CL, Vandervoort AA. Age-related changes in motor unit function. Muscle & nerve. 1997;20(6):679–690. [PubMed: 9149074]
31. Kugelberg E Adaptive transformation of rat soleus motor units during growth: Histochemistry and contraction speed. Journal of the neurological sciences. 1976;27(3):269–289. [PubMed: 131182]
32. Willadt S, Nash M, Slater C. Age-related changes in the structure and function of mammalian neuromuscular junctions. Annals of the New York Academy of Sciences. 2018;1412(1):41–53. [PubMed: 29291259]
33. Brown M, Holland R, Hopkins W. Motor nerve sprouting. Annual review of neuroscience. 1981;4(1):17–42.
34. Nockolds C, Hosker G, Kiff E. Fatigue rate of the external anal sphincter. Colorectal Disease. 2012;14(9):1095–1100. [PubMed: 22122610]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

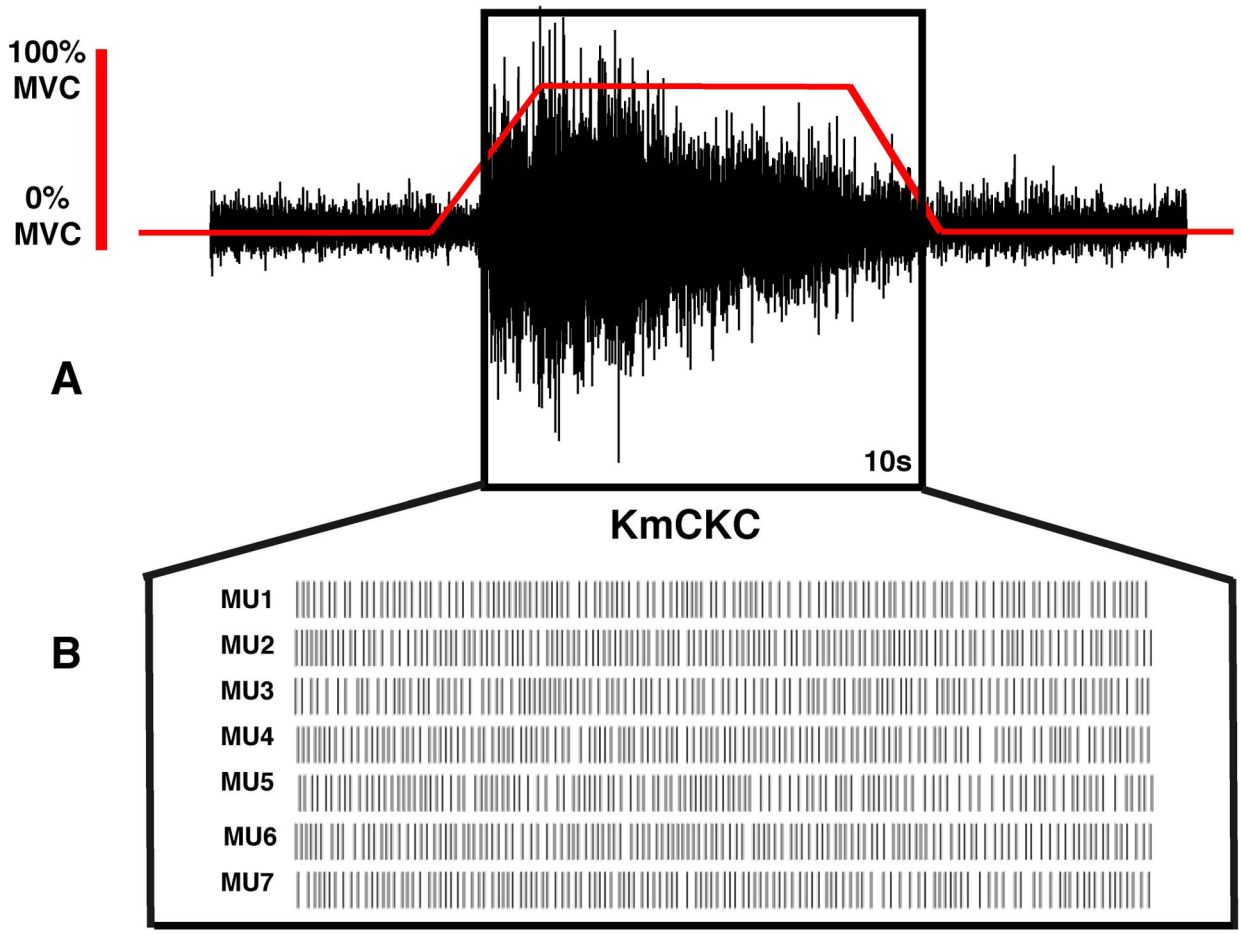


Figure 1:
(a) EMG Recording Paradigm; (b) Decomposed spike trains from the contraction segmented in (a).

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

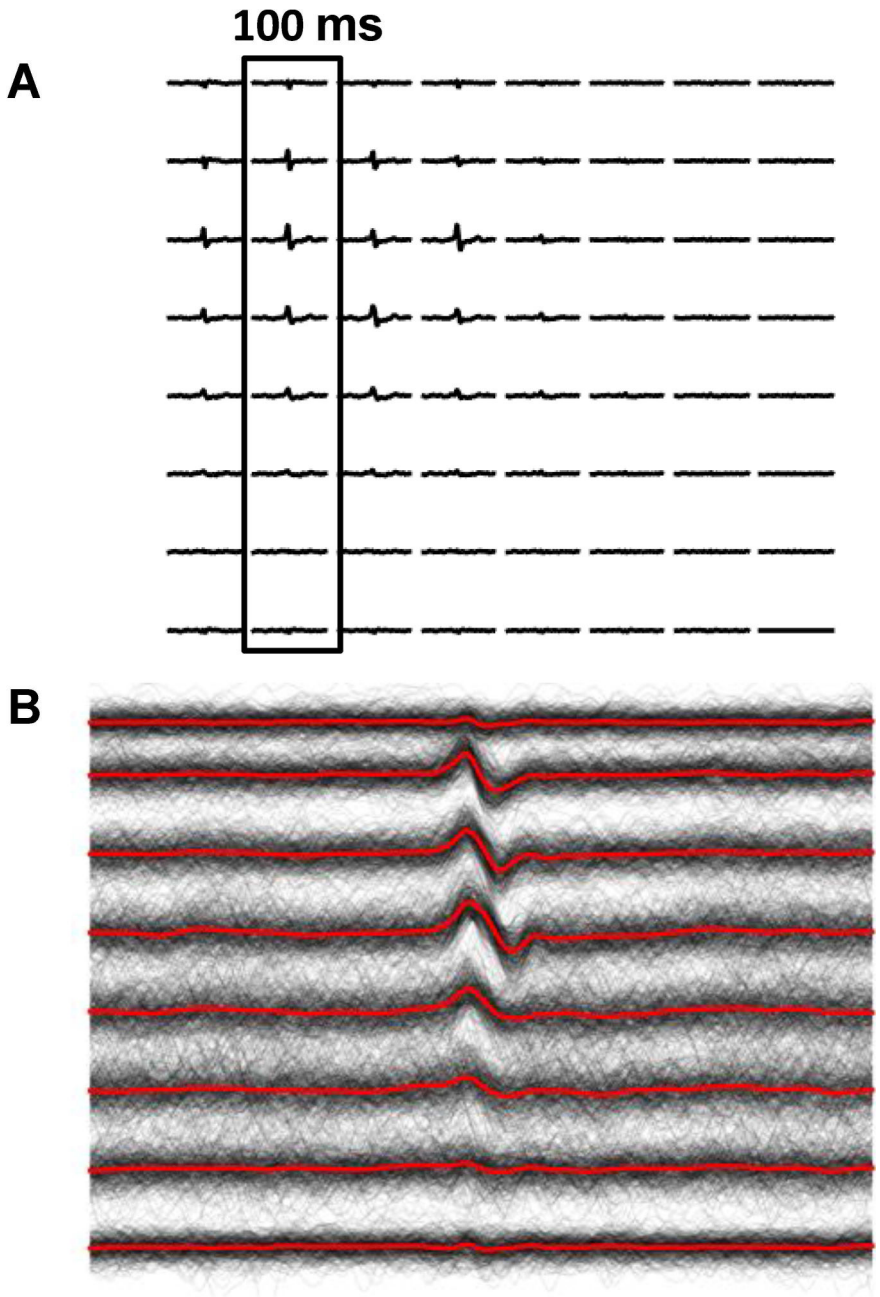


Figure 2:
(a) 64 channel HD-sEMG profile; (b) spike-triggered averaging results of MU1 from the selected column of (a); the black traces show the overlapping plot for all discharge instances, and the red traces shows the averaged MUAP waveform.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

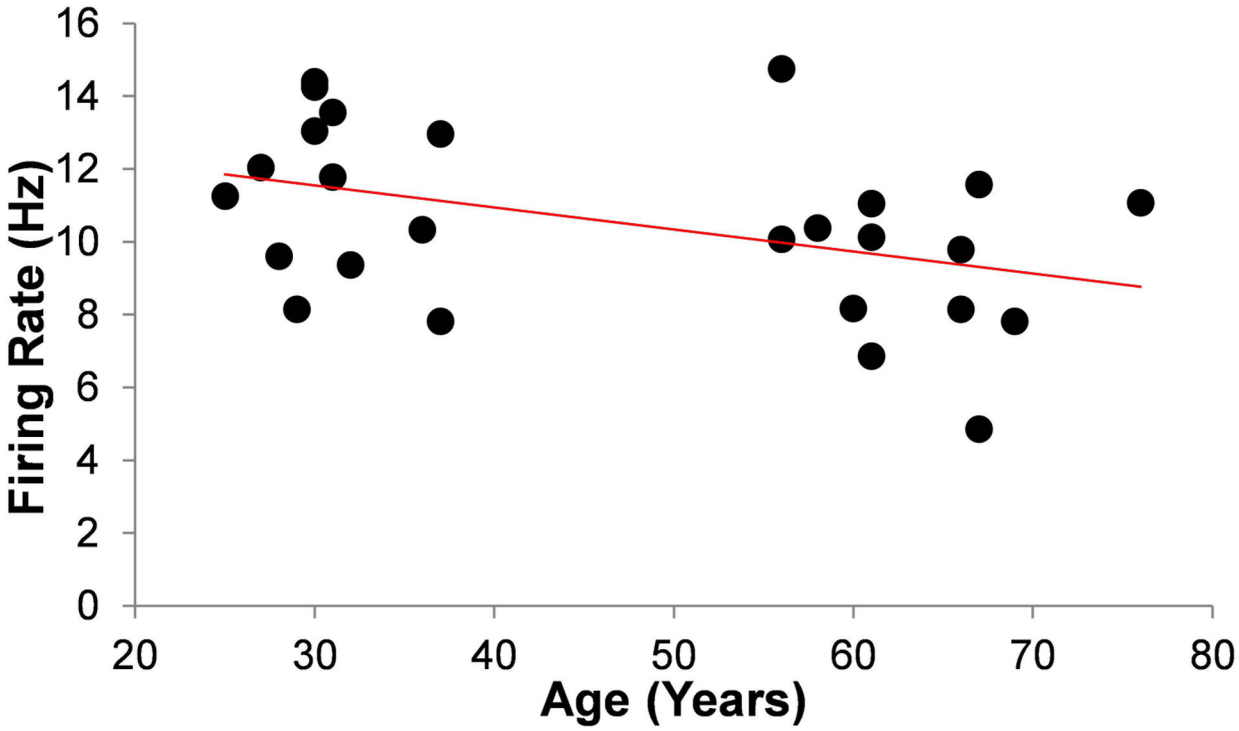


Figure 3:
Effect of aging on average MUAP firing rate.

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

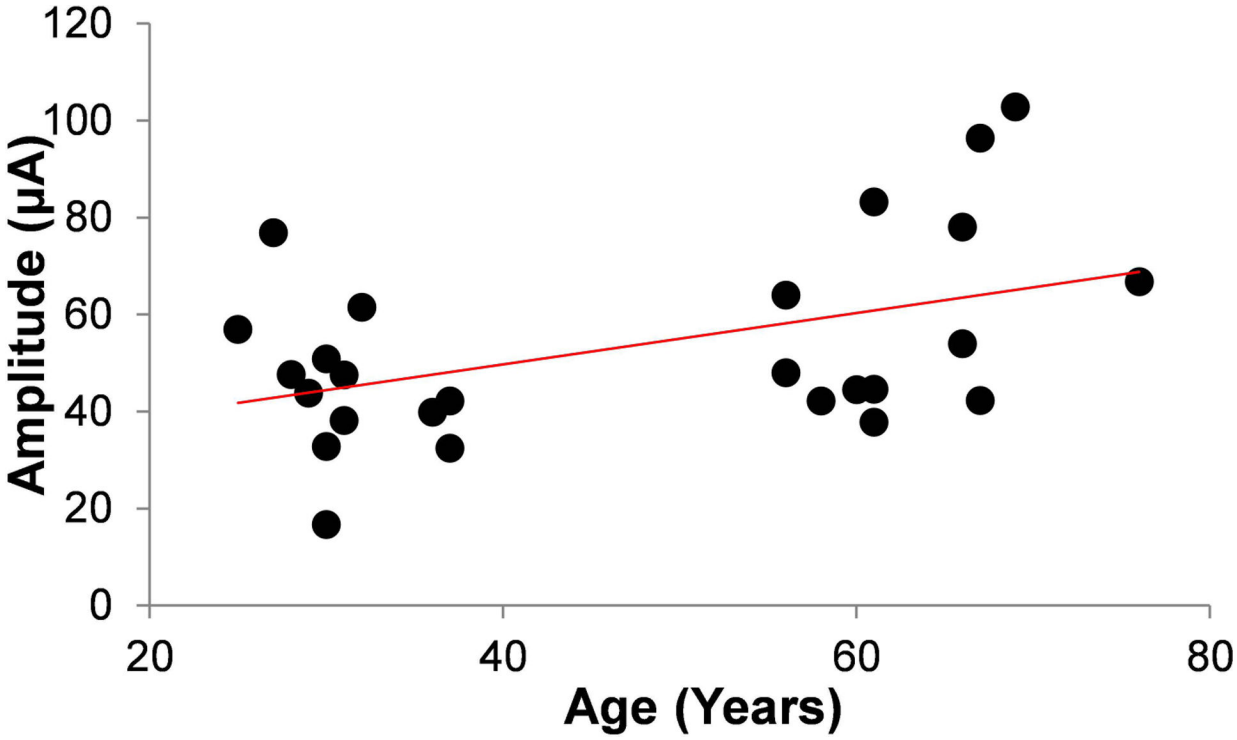


Figure 4:
Effect of aging on MUAP Amplitude.

Author Manuscript

Author Manuscript

Author Manuscript

Author Manuscript

Table 1

summarizes demographic information for both groups.

	Age (years)	Height (cm)	Weight(kg)	BMI(kg/m^2)	Parity
Young	31.0±3.6	158.6±3.4	52.6±5.0	20.9±2.3	1.5±0.5
Old	64.3±6.2	155.8±4.4	53.8±6.4	22.2±2.7	2.4±1.0